Mohamed Amer

AI Engineer | LLM & RAG Systems | LLMOps & AI Deployment

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience across Machine Learning, Computer Vision, NLP, Generative AI, RAG systems, AI agents, and LLMOps. Experienced in building end-to-end AI applications using Python, LangChain, LangGraph, CrewAI, FastAPI, Streamlit, vector databases, embeddings, Docker, CI/CD, and cloud deployment. Focused on developing reliable, scalable AI systems from experimentation through deployment and evaluation.

TECHNICAL SKILLS

ML & Deep Learning: Scikit-Learn, TensorFlow, Keras, Hugging Face Transformers, Model Training, Evaluation, Data Preprocessing
Computer Vision: OpenCV, MediaPipe, Image Processing, Feature Extraction, Object Detection, Image Segmentation, OCR/Tesseract
NLP: Text Preprocessing, Auto-Correction, NER, Sentiment Analysis, Text Summarization, spaCy, Transformers
LLM & Generative AI: Gemini, OpenAI, Groq, Prompt Engineering, LLM Application Development, RAG, Embeddings
Agentic AI: LangChain, LangGraph, CrewAI, Multi-Agent Systems, Tool/API Integration
RAG & Vector Search: FAISS, Pinecone, Retrieval Pipelines, Document Ingestion, Chunking, Ranking, Grounding
LLMOps & DevOps: Docker, Jenkins, CI/CD, LangSmith, Git, GitHub Actions, uv, pytest, Evaluation & Testing
AI Application Development: FastAPI, REST APIs, Streamlit, Flask, Real-Time Inference, ML/LLM Deployment
Cloud & Deployment: AWS ECS Fargate, Azure, Containerized Deployment, API Deployment
Data & Analysis: NumPy, Pandas, Matplotlib, Seaborn, Jupyter Notebooks

EXPERIENCE

Summer Intern (Flutter) – Information Technology Institute (ITI) | *Summer 2026*
Completed an intensive application-development internship focused on building applications using Flutter.

AI Intern – Route | *Feb 2024 – Jul 2024*
Built applied ML, DL, CV, NLP, and chatbot projects including object detection, sentiment analysis, text correction, summarization, face-mask detection, emotion recognition, and traffic-light detection.

Deep Learning for Computer Vision – University of Colorado Boulder | *May 2025 – Oct 2025*
Gained hands-on experience in object detection, segmentation, and intelligent computer-vision systems using deep learning architectures.

AI & Computer Vision Training – NTI | *Nov 2025 – Jan 2026*
Developed applied deep-learning and computer-vision models with hands-on experience in intelligent system development.

AI & Computer Vision Training – Huawei | *Aug 2024 – Oct 2024*
Trained on ML, DL, and computer-vision concepts through practical tasks and applied models.

AI & Deep Learning Training – Zewail City | *Jun 2023 – Nov 2023*
Developed Weather, Booking, Todo, and other applications while building foundational software and AI development skills.

EDUCATION

Bachelor's Degree – Artificial Intelligence, Faculty of Engineering, Mansoura University | *Sep 2022 – Jun 2027*
Artificial Intelligence undergraduate program.

SELECTED PROJECTS

Production RAG & LLMOps System
Built an end-to-end RAG application with FastAPI, LangChain LCEL, document ingestion, Google text embeddings, FAISS session indexes, Gemini/Groq LLMs, and LangSmith tracing/evaluation. Containerized the application with Docker and worked through automated testing and deployment workflows including Jenkins CI/CD and AWS ECS Fargate.

AI Document Intelligence RAG
Designed a document-intelligence workflow for PDF, DOCX, CSV, XLSX, and TXT sources, covering extraction, routing, retrieval/ranking, evidence qualification, provenance, calculations, and grounded answer generation. Explored production considerations including OCR, citations, RBAC, vector search, evaluation, latency, testing, and rollback.

LLMOps Automated CI/CD Pipeline
Configured a Jenkins CI/CD pipeline for an LLMOps repository using Python 3.12, uv, and pytest for automated code evaluation and testing, with Git-based development and deployment practices.

Multimodal Autism Spectrum Disorder Screening – Graduation Project

Developing a multimodal AI screening system combining images, audio, and questionnaire data to extract behavioral features and estimate ASD risk. Technologies include ResNet, OpenCV/MediaPipe, wav2vec 2.0, AssemblyAI, Gemini, scikit-learn, and Streamlit.

Project FORESIGHT

AI-Powered Multi-Document RAG Platform (LLMOps)

Developed a production-ready Generative AI application for interactive chat with multiple PDFs using Retrieval-Augmented Generation (RAG). Built a robust asynchronous API using FastAPI, LangChain, and FAISS. Integrated LangSmith for real-time pipeline tracing and quality control. Engineered an automated CI/CD pipeline with Jenkins and pytest. Successfully containerized and deployed the complete architecture to AWS, ensuring high availability, scalability, and seamless user experience.

AI Document Intelligence System with RAG

Developed an AI-powered Document Intelligence System that enables users to upload PDF, DOCX, XLSX, CSV, and TXT files and interact with them using natural language. Built with a Retrieval-Augmented Generation (RAG) pipeline, the system extracts content, generates vector embeddings, stores them in FAISS, and retrieves the most relevant context to deliver accurate, context-aware responses using an LLM. Designed with Streamlit for an intuitive user experience and scalable AI-powered document analysis.

TAIF – AI-Powered Multimodal Autism Screening System

TAIF is an AI-powered multimodal Autism Screening & Decision Support System designed to support early ASD risk assessment. The platform combines Computer Vision (fine-tuned ResNet50), Machine Learning questionnaire models, Speech Analysis, and Multimodal Fusion to classify autism risk as Low, Moderate, or High. It also provides AI-generated therapy recommendations, therapeutic stories, a RAG-based caregiver assistant, and a bilingual (Arabic/English) Streamlit interface. Built using Python, TensorFlow, Scikit-learn, OpenCV, Streamlit, and SQLite.

AI-Powered Face Detection System

Designed and implemented an AI-driven computer vision application for automated facial detection, analysis, and visual highlighting. The application processes group and individual image inputs, executes multi-step detection filtering, and displays precise bounding boxes through a scalable backend and clean web interface.